

Pranay Goel

+91 8368617372 | pranaygoel92@gmail.com | GitHub | LinkedIn | LeetCode

SUMMARY

Software Engineer and 2026 Computer Science graduate with production internship experience building **Python APIs, cloud-native backend services, data pipelines, and enterprise AI systems**. Strong foundation in **data structures, algorithms, object-oriented programming, databases, and system design**, with hands-on experience deploying software using AWS, Docker, and Kubernetes.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, JavaScript, SQL
- **Software Engineering:** Data Structures & Algorithms, OOP, System Design, RESTful APIs, microservices, Git, pytest
- **Backend & Messaging:** FastAPI, Node.js, API integrations, Apache Kafka, Amazon SQS
- **Databases & Storage:** PostgreSQL, DynamoDB, Redis, ClickHouse, Pinecone, HDFS, Parquet
- **Cloud & DevOps:** AWS (Lambda, S3, RDS, IAM, Secrets Manager, CloudWatch, Bedrock, EventBridge), Docker, Kubernetes, GitHub Actions (CI/CD), Argo CD, CloudFormation, AWS SAM
- **Data & Distributed Computing:** Apache Spark, PySpark, Spark SQL, dbt, ETL/ELT pipelines, pandas
- **AI Engineering:** LangGraph, OpenAI Agents SDK, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), tool calling, text-to-SQL, LLM evaluation

EXPERIENCE

- **Eli Lilly Services India Pvt. Ltd.** Jan 2026 – Jul 2026
Data Engineering Intern Bengaluru, India
 - Developed **Python REST APIs** using **AWS Lambda and DynamoDB** for Marketplace, Lilly's global data catalog serving **1,000+ users and 10,000+ data products**; delivered metadata, versioning, and governed workflows for access requests, approvals, provisioning, expiry, and status reconciliation.
 - Contributed to a Kubernetes-deployed **Python telemetry ingestion service** whose SDK batched events from 3 enterprise products into a REST collector for validation, deduplication, and asynchronous **ClickHouse** insertion.
 - Implemented reusable **Python SQL builders** for canonical telemetry metrics, keeping definitions consistent across services and agents; supported sub-second queries over millions of events powering **15+ dashboards for 140+ users**.
 - Built an in-process tool-calling loop in the **Python gateway** that retained the caller's token, invoked Marketplace APIs, and protected write operations through preview and confirmation.
 - Automated production deployments using **GitHub Actions, Docker, Kubernetes, and Argo CD**; integrated **Cloudflare WAF and Azure APIM** security controls and passed internal SecOps penetration testing.
 - Built a shared **DynamoDB conversation store** for both agents, providing persistent chat history and efficient retrieval of each user's previous sessions.
 - Built an agent for Marketplace using Anthropic Claude through Lilly's internal AI platform backed by **Amazon Bedrock**, making data discovery, access requests, status tracking, and approvals accessible through natural language and reducing user clarifications by **40%**.
 - Developed a Bedrock-powered **ClickHouse analytics agent** and dedicated **MCP server**, using deterministic tools for standard metrics and constrained read-only SQL generation for exploratory questions.
- **Netsmartz Infotech** May 2024 – Jul 2024
Data Analytics Intern Gurgaon, India
 - Transformed **500K-row datasets** with **Power Query** and DAX into analytics-ready reporting models.
 - Built **Power BI** dashboards that automated recurring reporting, eliminating **5+ hours/week** of manual work.

PROJECTS

- **Multi-Agent Assistant Platform** | *Python, Pinecone, JSON Schema, RAG, MCP* | GitHub
 - Designed a modular **multi-agent workflow** for request guarding, intent routing, RAG, and recommendations; consolidated guard and classification into one **structured-output call**, reducing **LLM calls per request by 40%**.
 - Implemented **Pinecone-backed RAG** and market-basket recommendations; built a provider-agnostic LLM client with JSON Schema routing and a **35-case evaluation harness** measuring routing accuracy, latency, and estimated cost.
- **Distributed Orbital Conjunction Analytics** | *Python, PySpark, Spark SQL, HDFS, Parquet* | GitHub
 - Engineered a distributed **PySpark** pipeline that parsed **15,487 TLE records** and propagated seven days of orbital position and velocity data using SGP4, storing date-partitioned Parquet datasets in HDFS.
 - Built close-approach detection using timestamp-aligned self-joins, 3D distance, relative velocity, and temporal clustering; identified **427 unique conjunction events in 13 minutes** on a three-worker Spark cluster.
 - Enriched conjunctions with SATCAT ownership metadata using **broadcast joins** and built a cached Streamlit/Plotly dashboard supporting local CSV/Parquet and HDFS-backed analytics.

EDUCATION

- **Manipal Institute of Technology** Manipal, India
Bachelor of Technology in Computer Science and Engineering, Minor in Big Data | CGPA: 8.23/10 2022 – 2026
 - **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Computer Networks, Object-Oriented Programming (OOP), System Design

CERTIFICATIONS

- **Big Data Specialization:** University of California San Diego, 2025
- **Deep Learning Specialization:** DeepLearning.AI, 2025 (Convolutional Neural Networks; CNNs in TensorFlow)